

Multiple Choice Questions

1st Question Group: operating system basics, structure and operatoin

1. The IEEE POSIX standardizes user and programming interfaces - True
2. The system libraries are running in protected mode - True
3. Modular kernels are able to load or unload kernel modules during runtime - True
4. The goal of the microkernel concept is to reduce the code which are running in protected mode, therefore the efficiency of the system is higher - False
5. During the boot process of a Unix system, the init process starts the user services - True

2nd question group: task management and scheduling

1. In common operating system no suspended state presents - False
2. The thread is a task which shares memory, stack and heap with other threads within the same process - **True but except the STACK, so the answer is FALSE**
3. The number of context changes of an I/O intensive task is typically higher than the number of context changes of a CPU intensive task on a lightly loaded system - True
4. The SJF scheduler has lower overhead when the ready to run task are stored in a tree data structure - FALSE
5. The convoy effect may occur when a FCFS scheduler is used - TRUE (convoy effect is when the task pile up when they will use the CPU for longer)

3rd question group: memory management

1. In an operations system typically there are more page tables than frame tables - TRUE
2. The number of page faults are in linear relation with the number of running tasks - FALSE
3. The CPU cannot access the data directly which are stored in the swap - TRUE
4. The malloc() system call allocates the whole requested memory section immediately - FALSE
5. With the copy-on-write method, it is possible for two tasks to access the same physical frames - TRUE

4th question group: file and storage system

1. The installed user programs on Windows 10 are located in the /Windows directory - FALSE
2. There is no need for backups if we are using a suitable redundant storage system, because it protects against all types of data loss - FALSE
3. A task can ignore file locking if we are using only advisory locking - TRUE (can ignore, just needs that the OS provides the synchronization for the tasks)

4. There is no data loss when 2 disk fails and we are using RAID6 - TRUE
5. With LVM (Logical Volume Management) we can create a file system with a size that doesn't fit on any physical storage system - TRUE